

INNER joins

```
CREATE DATABASE IF NOT EXISTS college_dbms;  
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;  
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (  
    Dept_ID INT PRIMARY KEY,  
    Dept_Name VARCHAR(50)  
);  
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Dept_ID INT,  
    Marks INT,  
    City VARCHAR(50)  
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES  
(10, 'CSE'),  
(20, 'ECE'),  
(30, 'MECH');
```

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES  
(101, 'Spoorthi', 10, 85, 'Delhi'),  
(102, 'Neha', 20, 92, 'Mumbai'),  
(103, 'Navya', 10, 76, 'Delhi'),  
(104, 'Manya', 30, 88, 'Bangalore'),  
(105, 'Tanisha', 20, 95, 'Delhi');
```

```
SELECT Student.Student_ID, Student.Name, Department.Dept_Name, Student.Marks  
FROM Student  
INNER JOIN Department  
ON Student.Dept_ID = Department.Dept_ID;
```

Result Grid					Filter Rows:	Export:	Wrap Cell Content:
Student_ID	Name	Dept_Name	Marks				
101	Spoorthi	CSE	85				
102	Neha	ECE	92				
103	Navya	CSE	76				
104	Manya	MECH	88				
105	Tanisha	ECE	95				

Result 9				
Output				
Action Output				
#	Time	Action	Message	
✓ 121	12:46:07	CREATE TABLE Student (Student_ID INT PRIMARY KEY, Name VARCHAR(50), Dept_ID INT, Marks IN...	0 row(s) affected	
✓ 122	12:46:07	INSERT INTO Department (Dept_ID, Dept_Name) VALUES (10, 'CSE'), (20, 'ECE'), (30, 'MECH')	3 row(s) affected Records: 3 Dupli	
✓ 123	12:46:07	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, 'Spoorthi', 10, 85, 'Delhi'), (102, 'Ne...	5 row(s) affected Records: 5 Dupli	
✓ 124	12:46:07	SELECT Student.Student_ID, Student.Name, Department.Dept_Name, Student.Marks FROM Student INNER JOIN ...	5 row(s) returned	

LEFT Join

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (
    Dept_ID INT PRIMARY KEY,
    Dept_Name VARCHAR(50)
);
```

```
CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Dept_ID INT,
    Marks INT,
    City VARCHAR(50)
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
```

(30, 'MECH');

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Spoorthi', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Navya', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');
```

```
SELECT Student.Name, Department.Dept_Name
FROM Student
LEFT JOIN Department
ON Student.Dept_ID = Department.Dept_ID;
```

The screenshot shows a database management tool interface. At the top, there's a 'Result Grid' section with a table containing 5 rows and 2 columns: 'Name' and 'Dept_Name'. The rows are: Spoorthi (CSE), Neha (ECE), Navya (CSE), Manya (MECH), and Tanisha (ECE). Below this is an 'Output' section with a dropdown menu set to 'Action Output'. This section displays a log of database actions with columns for '#', 'Time', 'Action', and 'Message'. The log shows four actions: creating a table (0 rows affected), inserting into the department table (3 rows affected), inserting into the student table (5 rows affected), and a select query (5 rows returned).

	Name	Dept_Name
▶	Spoorthi	CSE
	Neha	ECE
	Navya	CSE
	Manya	MECH
	Tanisha	ECE

#	Time	Action	Message
✓ 130	12:48:30	CREATE TABLE Student (Student_ID INT PRIMARY KEY, Name VARCHAR(50), Dept_ID INT, Marks IN...	0 row(s) affected
✓ 131	12:48:30	INSERT INTO Department (Dept_ID, Dept_Name) VALUES (10, 'CSE'), (20, 'ECE'), (30, 'MECH')	3 row(s) affected
✓ 132	12:48:30	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, 'Spoorthi', 10, 85, 'Delhi'), (102, 'Ne...	5 row(s) affected
✓ 133	12:48:30	SELECT Student.Name, Department.Dept_Name FROM Student LEFT JOIN Department ON Student.Dept_ID = De...	5 row(s) returned

RIGHT Join

```
CREATE DATABASE IF NOT EXISTS college_dbms;
USE college_dbms;
```

```
DROP TABLE IF EXISTS Student;
DROP TABLE IF EXISTS Department;
```

```
CREATE TABLE Department (
```

```
Dept_ID INT PRIMARY KEY,  
Dept_Name VARCHAR(50)  
);
```

```
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Name VARCHAR(50),  
    Dept_ID INT,  
    Marks INT,  
    City VARCHAR(50)  
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES  
(10, 'CSE'),  
(20, 'ECE'),  
(30, 'MECH');
```

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES  
(101, 'Spoorthi', 10, 85, 'Delhi'),  
(102, 'Neha', 20, 92, 'Mumbai'),  
(103, 'Navya', 10, 76, 'Delhi'),  
(104, 'Manya', 30, 88, 'Bangalore'),  
(105, 'Tanisha', 20, 95, 'Delhi');
```

```
SELECT Student.Name, Department.Dept_Name  
FROM Student  
RIGHT JOIN Department  
ON Student.Dept_ID = Department.Dept_ID;
```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Name	Dept_Name
▶	Navya	CSE
	Spoorthi	CSE
	Tanisha	ECE
	Neha	ECE
	Manya	MECH

Result 11

×

Output

Action Output

▼

#	Time	Action	Message
✓ 139	12:49:45	CREATE TABLE Student (Student_ID INT PRIMARY KEY, Name VARCHAR(50), Dept_ID INT, Marks IN...	0 row(s) affected
✓ 140	12:49:45	INSERT INTO Department (Dept_ID, Dept_Name) VALUES (10, 'CSE'), (20, 'ECE'), (30, 'MECH')	3 row(s) affected Reco
✓ 141	12:49:45	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, 'Spoorthi', 10, 85, 'Delhi'), (102, 'Ne...	5 row(s) affected Reco
✓ 142	12:49:45	SELECT Student.Name, Department.Dept_Name FROM Student RIGHT JOIN Department ON Student.Dept_ID = D...	5 row(s) returned

FULL OUTER Join

CREATE DATABASE IF NOT EXISTS college_dbms;

USE college_dbms;

DROP TABLE IF EXISTS Student;

DROP TABLE IF EXISTS Department;

```
CREATE TABLE Department (
    Dept_ID INT PRIMARY KEY,
    Dept_Name VARCHAR(50)
);
```

```
CREATE TABLE Student (
    Student_ID INT PRIMARY KEY,
    Name VARCHAR(50),
    Dept_ID INT,
    Marks INT,
    City VARCHAR(50)
);
```

```
INSERT INTO Department (Dept_ID, Dept_Name) VALUES
(10, 'CSE'),
(20, 'ECE'),
```

(30, 'MECH');

```
INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES
(101, 'Spoorthi', 10, 85, 'Delhi'),
(102, 'Neha', 20, 92, 'Mumbai'),
(103, 'Navya', 10, 76, 'Delhi'),
(104, 'Manya', 30, 88, 'Bangalore'),
(105, 'Tanisha', 20, 95, 'Delhi');
```

```
SELECT Student.Name, Department.Dept_Name
FROM Student
LEFT JOIN Department ON Student.Dept_ID = Department.Dept_ID
```

UNION

```
SELECT Student.Name, Department.Dept_Name
FROM Student
RIGHT JOIN Department ON Student.Dept_ID = Department.Dept_ID;
```

Result Grid					Filter Rows:	Export:	Wrap Cell Content:
	Name	Dept_Name					
▶	Spoorthi	CSE					
	Neha	ECE					
	Navya	CSE					
	Manya	MECH					
	Tanisha	ECE					

Result 12 ×				
Output				
Action Output				
#	Time	Action		Message
✓ 148	12:51:36	CREATE TABLE Student (Student_ID INT PRIMARY KEY, Name VARCHAR(50), Dept_ID INT, Marks IN...		0 row(s) affected
✓ 149	12:51:36	INSERT INTO Department (Dept_ID, Dept_Name) VALUES (10, 'CSE'), (20, 'ECE'), (30, 'MECH')		3 row(s) affected Records
✓ 150	12:51:36	INSERT INTO Student (Student_ID, Name, Dept_ID, Marks, City) VALUES (101, 'Spoorthi', 10, 85, 'Delhi'), (102, 'Ne...		5 row(s) affected Records
✓ 151	12:51:36	SELECT Student.Name, Department.Dept_Name FROM Student LEFT JOIN Department ON Student.Dept_ID = De...		5 row(s) returned